

Table of Contents

Acknowledgement.....	1
Abstract.....	2
List of Abbreviations.....	2
Introduction.....	3-6
Background of the Study.....	3
Objective of the Study.....	5
Rationale of the Study.....	5
Scope and Limitations of the Study.....	5
Literature Review.....	6-7
Methodology.....	7-10
Topic Selection.....	7
Literature Review.....	8
Objectives Formulation.....	8
Data Collection.....	8
Data Analysis.....	8
Identification of Accident Hotspot and its Change over Years	8
Determine Accident Severity along N1.....	9
Identification of Critical Areas.....	9

Preparation of Land Cover Map for Critical Areas.....	9
Comparison of Different Trend to Analyse Relationship.....	10
Data Analysis.....	10-25
Accident Hotspot Analysis.....	10
Severity Analysis.....	13
Land Cover Change Analysis of Critical Areas.....	14
Comparison of Trends and Relation Analysis.....	23
Findings.....	25-26
Conclusion.....	26-27
Reference.....	27
Appendix	28

Identifying Hotspot and Severity of Road Accident along National Highway (N1) and Analyse the Influence of Corresponding Land Cover

Course: PLAN 362

Course Title: GIS and Remote Sensing Studio

Submitted to

Sadia Afroj

Assistant Professor, Department of URP, BUET

Md.Waliullah

Lecturer, Department of URP, BUET

Submitted by

Group 07

1915013 (Md. Ahnaf Nafi)

1915016 (Fariha Tanzim Deea)

1915024 (Rumkey Chowdhury)



Department of Urban and Regional Planning

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Acknowledgement

First, we would like to express our deepest gratitude to Almighty for giving us the strength and the composure to finish the report. We would like to express our deepest gratitude to all my course teachers, Sadia Afroj, Assistant Professor, Department of URP, BUET and Md.Waliullah, Lecturer, Department of URP, BUET; for their constant guidance,suggestion and encouragement in making of this project.

The journey throughout the study on “Identifying Hotspot and Severity of Road Accidents along National Highway (N1) and Analyze the Influence of Corresponding Land Cover” was great and compiling the report was a memorable experience for each and every one of us. The report has allowed us to test our strengths and weaknesses as a team. It was a pleasure to work together for all of us.

The study has been completed with the support of many people. So, we want to take the opportunity to acknowledge the generous help extended to us in our work. We would like to acknowledge Niaz Mahmud Zafri, Assistant Professor, Department of URP, BUET for providing us suggestions and answering some our questions in order to complete this project properly.

We also want to thank our seniors Iffat Baki Bushra and Nazifa Anzum apu , Department of Urban and Regional Planning of 17 batch for providing us advises and the data that has been collected from ARI.

Doing the study without the help of our classmates was difficult, our deepest gratitude to them. We have not enough valuable words to express our heartfelt thanks, but our heart is still full of the favours received from every person.

Abstract

The National Highway 1(N1) is 465 kilometers in length linking the country's two largest cities, Dhaka and Chittagong specifically Mongla port. For this reason, N1 is referred as backbone for the economy and main transportation artery in Bangladesh. Unfortunately, N1 is one of the most accident-prone highways in Bangladesh. To cater to this aspect, we have tried to identify some critical areas based on the total number of road accidents and severity of those consequences. Though in general highways are established far away for city center, N1 is different as it has crossed through the most vibrant areas of Chittagong city. To explore the influence of built-up area on road accident as built-up area is a common characteristics of city center, we have tried to analyze the land cover change over the years. To accomplish this, we have resorted to hotspot analysis in terms of total accident through Kernel Density, severity indexing, prepared land cover maps of different years for critical areas by supervised classification method after going through a number of GIS operations. After analyzing the maps, we have found different types of trends for hotspot, severity and land cover change (transition from non-built up to built up). For Comilla, Feni and Cox's Bazar; the increasing, fluctuating and constant hotspot trend and also the high, medium and low severity trend do not match with the trend of decreasing, increasing and fluctuating land cover change respectively. On contrary, fluctuating hotspot trend and high severity precisely match with fluctuating land cover change for Chittagong like increasing trend at initial to gradually decreasing trend in the same time period. Based on our study, we cannot clearly define that land cover has a relation with accident hotspot and severity as there are some exceptions also. To identify the exact association, further analysis is needed in future.

List of Abbreviations

RTA - Road Traffic Accidents

N1 – National Highway 1

GIS – Geographic Information System

ARI - Accident Research Institute

KDE - Kernel Density Estimation

NDBI- Normalized Difference Built-up Index

Introduction

Background of the Study

The issue of Road Traffic Accidents (RTAs) is a complex problem worldwide, with approximately 1.3 million fatalities and 50 million injuries resulting from RTAs each year globally (ITF, 2017). It is crucial to understand where and when accidents frequently occur to significantly reduce the number of accidents. These locations, which have a high incidence of accidents compared to other sites, are commonly referred to as hotspots or black spots (Dereli & Erdogan, 2017). Studies have demonstrated that the occurrences of RTAs are not random and are influenced by several key factors, such as geometric design, traffic volume, surroundings, and severe weather conditions (Xia & Yan, 2008). Therefore, to develop effective accident prevention strategies, it is imperative to identify potential hazardous locations and the associated accident occurrence times (Harirforoush, 2017). The KDE method has been commonly applied for identifying severe accident risk locations on road networks (Harirforoush & Bellalite, 2016; Vemulapalli et al., 2017).

Bangladesh is a densely populated, low-lying country with over 140 million inhabitants residing in 147,570 square kilometres. The National N1 Highway in Bangladesh is a vital transport artery that connects the country's capital city, Dhaka, with the port city of Chittagong. Bangladesh experiences roughly 5,000 reported police accidents yearly. The Dhaka-Chittagong highway is recognized as the second most accident-prone highway among Bangladesh's national highways. The N1 highway has significantly improved transportation between Dhaka and Chittagong, making it faster, safer, and more convenient for commuters and businesses. Traffic congestion on the

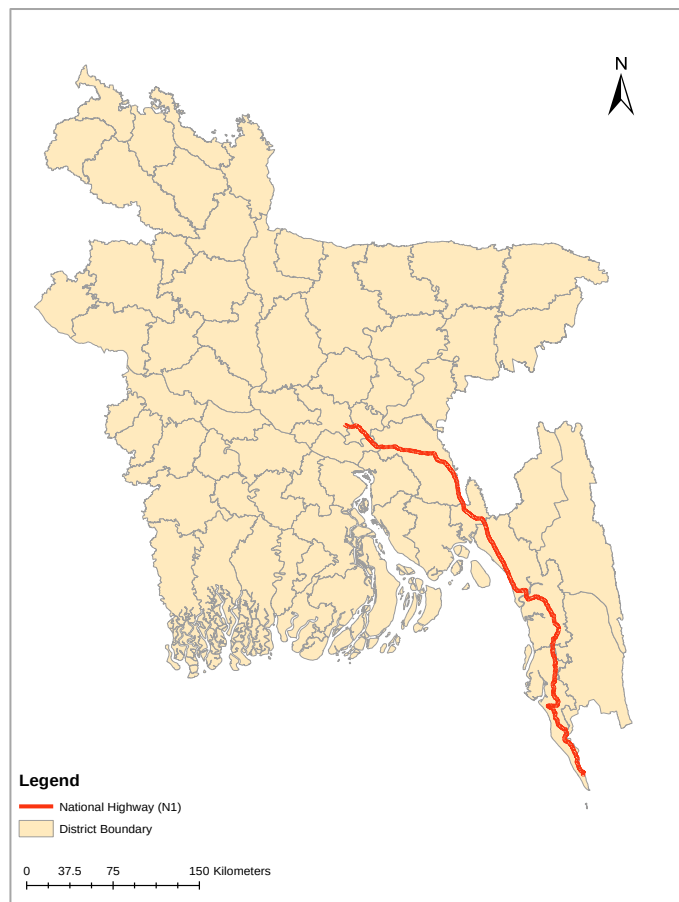


Figure 1: National Highway (N1)

N1 highway is a significant problem, leading to delays and reduced travel speeds. Inadequate capacity, poor geometric design, and lack of traffic management are significant causes of congestion. Road safety is another major issue on the N1 highway. Accidents and fatalities are common, often caused by speeding, reckless driving, and poor road conditions. Traffic congestion, road safety, and environmental impact will be essential to ensure the continued success of the highway as a vital transportation link in Bangladesh. (Saleh and Islam, n.d.).

A hotspot is a location or area where a high concentration of traffic accidents exists. A hotspot is a place on the road with an increased risk for traffic accidents, and the likelihood of such accidents is connected to the risk level in the neighboring regions. (Sharraf et. al., 2018). Spatial analysis techniques were used to identify hotspots based on the frequency and severity of accidents in specific locations. The hotspots were visualized on a map, which can be used to inform targeted interventions and policy decisions aimed at reducing traffic accidents in the city. GIS and spatial analysis tools can identify hotspots and prioritize areas for traffic safety improvements. By identifying and addressing hotspots, traffic safety measures can be implemented effectively and efficiently, reducing traffic accidents and associated injuries and fatalities. (Shafabakhsh et. al., 2017)

Accident severity indicates the seriousness of accident, or the level of damage occurred due to accident whether it is fatal/grievous like death, serious injury or non-fatal like slight injury, minor property damage. To measure the severity a method called severity indexing is applied where the type of accidents in terms of their severity is given a weightage and calculate a fraction number or index value for each observation. Based on that that value it is decided whether the accident was severe or not (Le et. al., 2019).

Land cover refers to the physical and biological cover on the Earth's surface, including natural features such as forests, grasslands, and water bodies and artificial features such as urban areas and agricultural land. The type of land cover can impact road safety. Rural areas with open fields and few natural barriers may have higher rates of severe accidents involving high speeds and loss of vehicle control. On the other hand, areas with dense forests may have a higher risk of collisions with wildlife. Changes in land use and land cover can also have an impact on road safety. For instance, the conversion of agricultural land to urban use may result in an increase in traffic volume and higher accident rates. Road safety and the frequency of road accidents can be affected by the land cover type and land use in an area (Lambin et. al., 2001).

Objectives of the study:

The study has been conducted to achieve the following objectives:

- To identify accident hotspots along national highway (N1) and find out critical areas based on change in accident hotspot over years incorporating accident severity.
- To observe land cover change of the critical areas and analyse the relation with accident.

Rationale of the study

Road accidents, which cause harm and deaths, are the most damaging impacts of developing a modern transportation system. At the global level, road traffic accidents are the seventh most common reason for death (WHO,2013). In context of Bangladesh, national highways account for a much more significant percentage of accidents and fatalities when compared to other known locations. Among the national highways, N1 is the main transportation artery in Bangladesh linking the country's two largest cities, Dhaka and Chittagong. It is the busiest of all the major highways in Bangladesh. As Mongla port is connected through this highway, N1 is referred backbone for the economy of Bangladesh. Moreover, N1 is one of the most accident-prone highways in Bangladesh. Generally, highways connect two cities and are built far away from the city center. A city center is mostly built-up based and accommodates activities like significant commerce, political, cultural etc. N1 has crossed through the most vibrant areas of Chittagong city. Moreover, N1 is considered as one of the most accident-prone highways in Bangladesh. So, there is potential of studying accident condition of this highway.

Due to rapid urbanization built up area has been increasing rapidly. In fact, the whole land cover is being changed. Different studies show that built environment has an influence on mobility and safety on road. Considering all these this study is meant to identify the most vulnerable areas along the nation highway 1 (N1) for road accidents and analyze whether there is any relation between land cover of those critical areas and accident in terms of severity and number.

Scope and Limitations of the Study

The main aim of this study is to identify the critical areas along National highway 1 (N1) in terms of total accident and severity. Some areas are accident-prone, but their consequences are not major, or accident rate is low but severe. On the other hand, some areas are critical for

accidents and consequences. Moreover, land cover can be an influencing factor on accident rate and severity. We cannot surely define that land cover has a strong relationship with being a hotspot for accident and severity as we have not found many studies regarding this topic. This study will try to connect three facts that are severity, hotspots based on total accidents and land cover with respect to identifying their trends in critical areas. But further examinations may be needed in future to reach a final decision.

The first limitation of the study is the lack of resources. We have not found many studies regarding these topics together especially for land cover change. Some studies have been conducted incorporating land use but for a large scale area like district detailed land use data is difficult to collect. The second limitation is the supervise method to create a land cover map in GIS. The reflectance of barren land and built-up area is almost the same. So, it is difficult to distinguish between them and the maximum time they are mixed up. The supervise method is not suitable for large study areas. We have found many errors in generated maps, and it is difficult to revise the errors in multiple maps.

The third limitation is this study needs future processing to reach a decision but in our study scale, we clearly cannot define if land cover has any significant relationship with accident hotspots.

Literature Review

A study in London developed a technique for examining the patterns of injuring incidents using a geographic information system (GIS) and an evaluation of Kernel density. This study aimed to offer a technique for locating accident hotspots utilizing GIS, Kernel density estimation, addition of accident-associated area data, and identification of connected regions using average K algorithm. In order to develop this approach and clustering technique, data on incidents that happened in the research region in London during the previous five years were examined. Moreover, this study often explores hotspots, clusters these sites, and applies semiotics based on spatial indices to these places (Anderson, 2009).

One of the most crucial issues facing traffic safety engineers worldwide is where to deploy safety precautions so that they may have the greatest influence on traffic safety. Sites on a portion of roads or highways with a greater accident frequency than predicted at some threshold

levels of importance are referred to as "Hot Spots," "Black Spots," or "High Accident Locations." (Choudhary, Ohri, & Kumar, 2015). Road repair surrounding downtown historical sites was made possible by this study. A study was conducted in Turkey to assess the distribution of accidents on a highway near the city of Afyonkarahisar's entrance utilised (Erdogan et al., 2008). From a study that uses GIS and statistical analysis to identify road traffic accident hotspots in Hanoi, Vietnam. The authors employ various temporal and spatial methods to identify high-risk areas and assess the contributing factors to accidents. The study highlights the importance of utilizing spatial analysis techniques to better understand the spatial distribution and patterns of road traffic accidents, and to inform effective road safety interventions in Hanoi, Vietnam (Le, K. G., Liu, P., & Lin, L. T., 2020).

A study conducted in the United States found that land use mix, which measures the diversity of land uses in a given area, was positively associated with the incidence of pedestrian and bicycle accidents (Ewing et al., 2013). Another study of Malaysia North-South Expressway illustrates that accidents tend to occur where continuous traffic flow is disturbed (Norhafizah et al., 2019). Research has been conducted in several countries, such as China, Thailand, the United States, Pakistan, and India, to study how the landcover along roadways has changed over time (Lein & Day, 2008). However, no similar studies have been conducted in Bangladesh. Nevertheless, no studies have been conducted on major highways in Bangladesh. Although several studies have investigated the frequency of accidents along various national routes in Bangladesh, none have explored the connection between roadway accidents and landcover change.

Methodology

Topic Identification

Road accident has become one of the most concerning issues of modern life. Keeping that in mind several literatures has been reviewed to identify a suitable topic for the study relevant to GIS application. Several studies have been on road accident specifically incorporating spatial analysis tool GIS. Reviewing all those literatures it become clear that road accident has some distribution patter over space which is identified through hotspot analysis. On the basis of this the study topic has been selected to identify accident hotspots along national highway N1 and determine whether there is any relationship with landcover.

Literature Review

After the selection of study topic further some literatures have been reviewed to get idea about different methods of accident hotspot analysis, accident severity indexing etc. Some literature also refer significant landcover change for different buffer distance of highways. All those help to construct the methodology as well as the analysis of study.

Objective Formulation

Every study needs to be focused on certain goals. These focal points are the objectives. Two major objectives have been formulated to conduct the study which is mentioned in the introduction chapter. The entire process of this study has been established on the basis of these objectives.

Data Collection

For the purposes of the study relevant data has been collected which is from secondary sources. Collected data with their respective sources are given below.

Table 1: Data type with sources

Data	Source
Number of total accidents per unit length of National Highway(N1)	Accident Research Institute (ARI), BUET
Accident Location (District)	
Accident Severity	
Land Cover	Landsat-4-5,8

(Source: Author, 2023)

Data Analysis

After the collection of data, it has been primarily processed with the help of Microsoft Excel to get structured properly for further analysis in GIS.

Identification of Accident Hotspot and its Change over Years

Data collected from ARI, shows total number of accidents per unit length of road (N1) starting from Teknaf. Using this data as input we get several hotspots and cold spots of accident through Kernel Density Estimation which is an inbuilt spatial analyst tool of Arc GIS. Kernel density estimation (KDE) is a non-parametric method used to estimate the probability density function of a random variable. The importance of kernel density estimation lies in its ability to provide

a smooth estimate of the underlying density of a data set, even when the data is multi-modal or has a complex structure (Anderson, 2007). At first a 1km buffer around the national highway N1 has been created as literature shows different buffer range from highway within which the influence of built-up area seems to be significant. For each interval of years, the hotspots are identified and compare to detect the change in accident pattern over years to identify trends.

Determination of Accident Severity along N1

Severity indicates the fatality of accident whether the accident is fatal like death or non-fatal like slight injury. Severity related data includes different types such as fatal, grievous, minor property damage. To identify unsafe locations, accidents should be weighted according to severity. To take severity of accidents in consideration, this study employs a severity weighting system alongside with incidental accident data in analysis of high and low clustering.

$$SI = 3 \times X_1 + 1.8 \times X_2 + 1.3 X_3$$

where: X_1 = total number of fatal accidents, X_2 = total number of grievous accidents, X_3 = total number of minor and property damage only accidents (Choudhary et. al., 2015).

Identification of Critical Areas

Observing the change in accident hotspot some trends are identified and along with that severity index has been incorporated to identify some critical areas on the basis of which rest of the analysis will proceed. Initially all the areas (District) along N1 seem to be the study area which later scaled down to some specific areas (District) through hotspot changing trend and severity indexing.

Preparation of Land Cover Map for Critical Areas

From the preceding step some critical areas specifically some districts through which the national highway N1 passes has been identified. Among those districts landcover map has been prepared for each one of different years over which accident hotspot and severity has been changed. Landcover maps have prepared using supervised classification (maximum likelihood method). For each feature class almost 70-80 training sample has been considered. From the year-to-year change in landcover we get a trend of transition from non-built up to built up area.

Comparison of Different Trend to Analyse Relationship

Through different several different analyses some trend has been identified hotspot changing trend, severity trend and landcover changing trend (transition trend from non-built up to built up). Putting all of these trends together in tabular form a comparison can be made and get some idea about their interrelation.

This study has been done while following the aforementioned methodology to reveal the findings and the analyzed data collected from secondary sources.

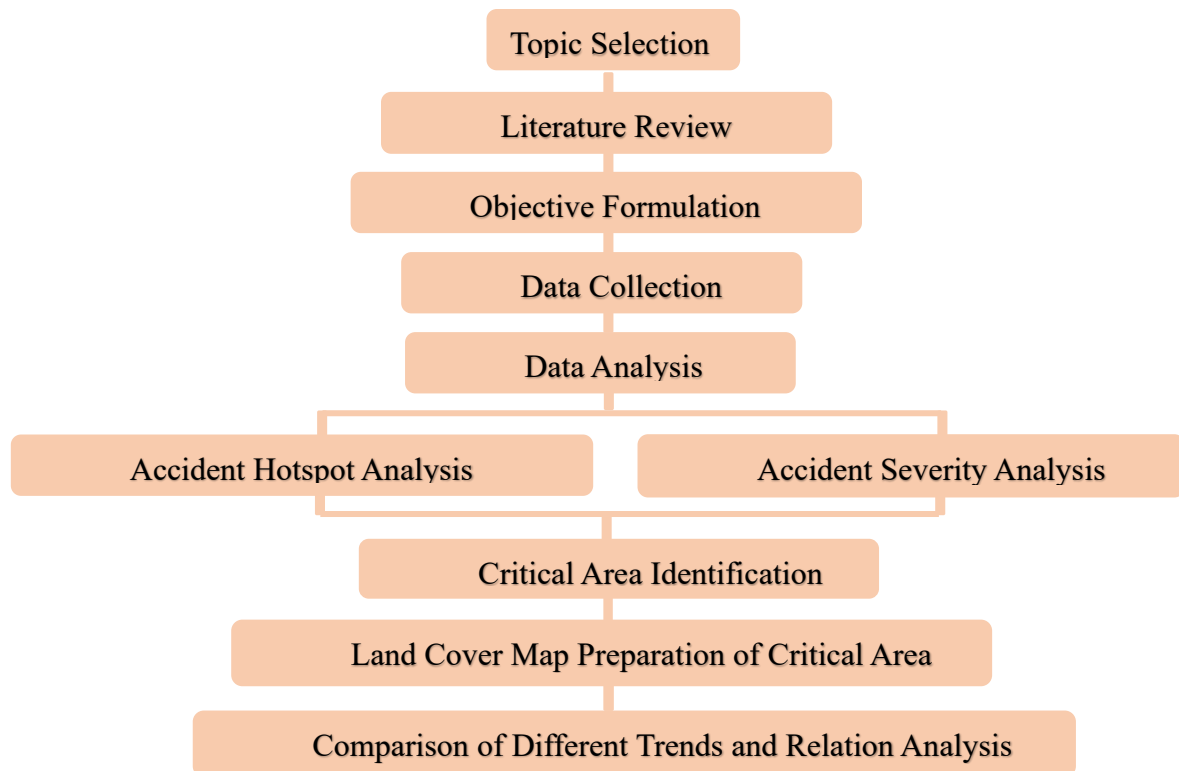


Figure 2: Flow diagram of the entire study process

(Source: Author, 2023)

Data Analysis

Accident Hotspot Analysis

We have found the information for the number of accidents in each length of the highway. We have taken 1km buffer zone from highway. According to literature review, the influence of built area is significant within 200m distance range from highway (Feng et al.,2021). The range of the distance differs in different country. For this reason, we have taken 1 km as an average buffer distance. The dataset on number of accidents for each kilometer road length is used to

analyze the hotspot through kernel density. Kernel Density estimation is unique compared to other estimations methods because it allows for the modeling the density using a flexible kernel function. The bandwidth of the kernel function can be adjusted to control the smoothness of the estimated density. Kernel density provides a smooth transition of low to high value. We can analyze which area is at high risk for accidents and which area is not very vibrant. Over the years, the hotspots and cold spots also evolve. Here we have tried to figure out the trend of changing hotspots. Chittagong, Feni and Comilla have shown drastic changes over the years. On contrary, Cox's Bazar is constantly identified as a cold spot area. For this reason, we have given our focus mainly on these four areas.

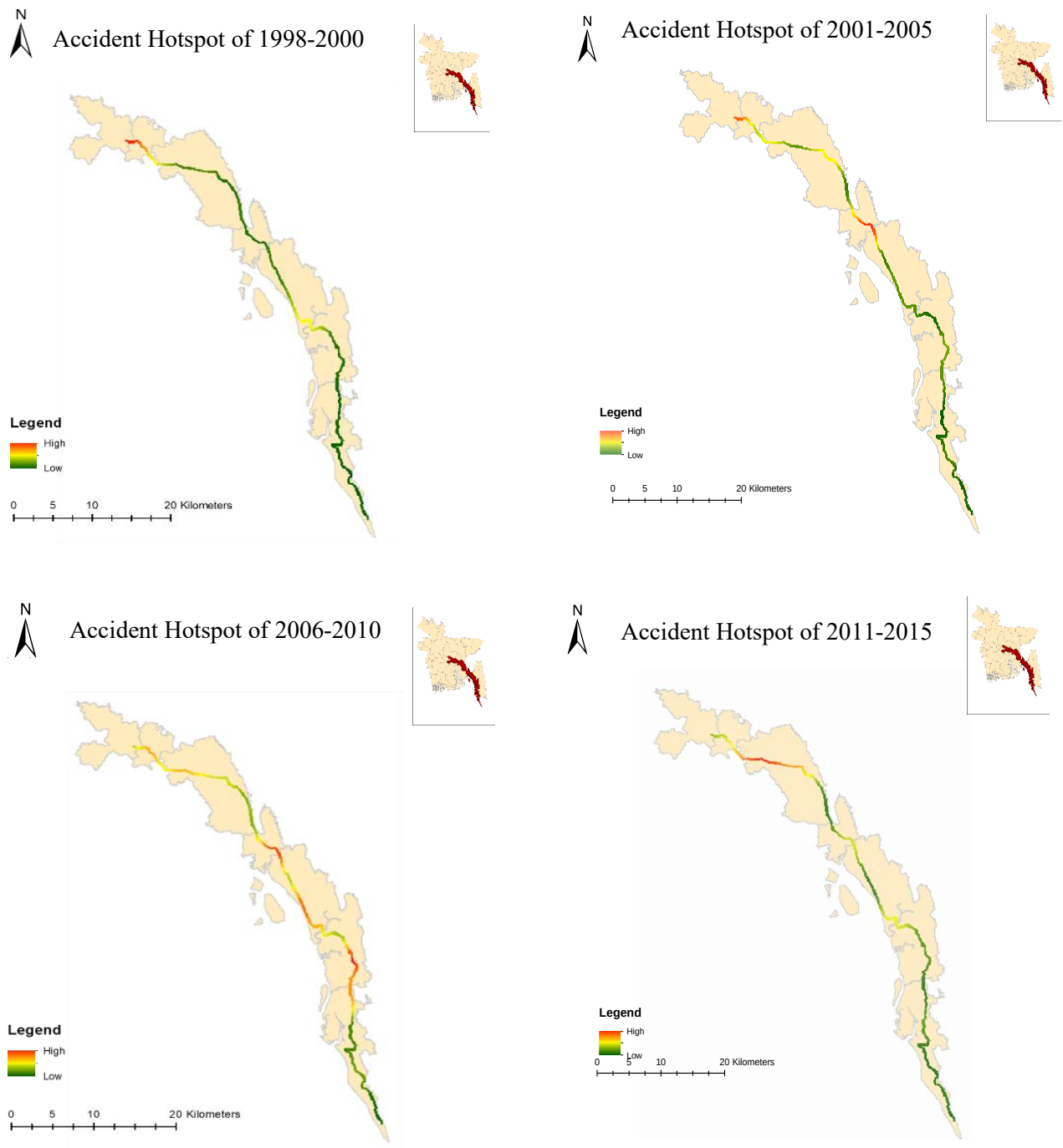
By analyzing the hotspots of different years, we have found three trends over the years. Chittagong and Feni have fluctuated in different years, but Comilla has constantly shown increasing trend of road accidents. Only Cox's Bazar is constant over the years.

In 1998-2000, Chittagong was mainly cold spot but slightly mild risk in Bakali and bayejid Bostami upazila at the middle of Chittagong. Within 2001–2005, these areas were becoming low risky whereas Fatikchhari became at high risk. In 2006-2010, whole Chittagong area became an intense hotspot area specially Fatikchhari, Hathazari, Banshakhali, and Anowara upazilas and Boalkhali was at average risk. On the other hand, in 2011-2015, suddenly whole Chittagong district turned into low-risk zone. Some areas under Bakali and Boalkhali upazilas were at mild risk.

In 1998-2000, Comilla was prominently in a low-risk zone. Some upazilas like Chaudhagram, Comila Sadar Dakshin were at lowest risk zone whereas Meghna was at slightly risk. In 2001-2005, Comila Sadar Dakshin, Meghna and Comilla Adarsha Sadar and Meghna upazila became average risk zone whereas other areas were at lowest risk zone. In 2006-2010, Dabedumedi became the most intense risk zone in Comila. Other upazilas were gradually changing into successive risk zone. In 2011-2015, half of Comilla district became highly risky specially Meghna to Burichang and other areas were at low-risk zone.

In 1998-2000, whole Feni district was under lowest risk zone. In 2001-2005, suddenly whole Feni district fall under highly risk zone. In 2006-2010, Feni was consistently at the same high-risk zone. In 2011-2015, Feni started to revive and whole district turned into very low risk zone. In 1998-2000, whole Cox's Bazar was at lowest risk zone. In 2001-2005, Cox's Bazar remained consistently lowest risk zone. In 2006-2010, Chakaria upazila suddenly turned into

high-risk zone and other upazilas remained same. In 2011-2015, whole Cox's Bazar again went back to its initial lowest risk zone.

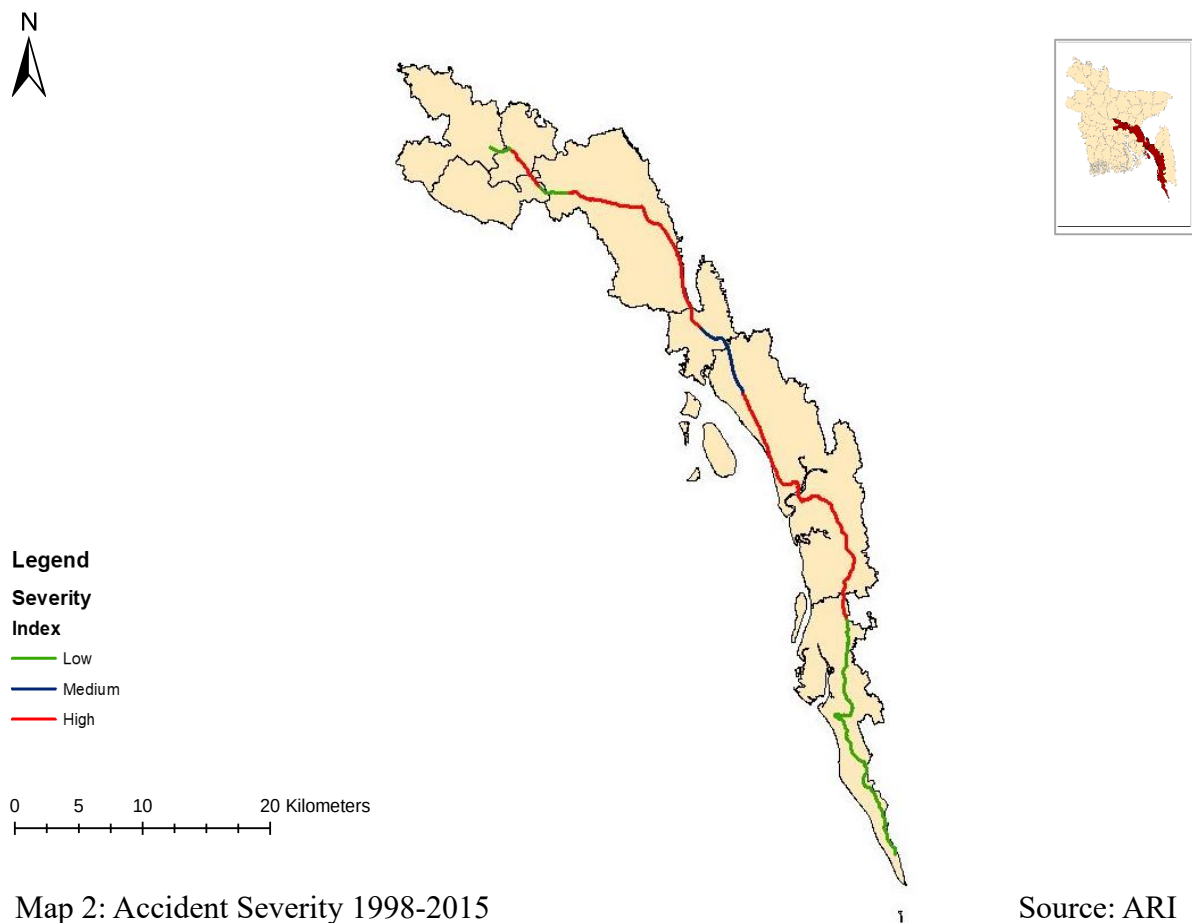


Map 1: Change in Accident Hotspots over years

Source: ARI

Accident Severity Analysis

The severity index includes various consequences of accidents. Here we have calculated the severity index integrating consequences like death, grievance and minor injuries. SI index shows different results from hotspots based on total no of accidents. Munshiganj was consistently at high-risk in terms of both severity and hotspot. It implies that the total number of accidents is high, and their consequences are also much severe. Naranganj was always at high risk for accidents, but their consequences were not very serious. In Comilla, the number of total accidents was gradually increased, and severity was high. It implies that the number of accidents had increased day by day and their consequences were also getting very serious. Feni showed fluctuations in terms of total accidents. Moreover, the consequences of accidents were moderate and serious. It implies that though the number of total accidents differed over time, their severity remained high. Chittagong had also shown the same trend of fluctuations in terms of total accidents and severity. Cox's Bazar was consistently in a low-risk zone of accidents. Moreover, the consequences of accidents were not very serious. It implies Cox's Bazar was the safest district in terms of accidents and severity of their consequences.



Land Cover Change Analysis of Critical Areas:

Based on hotspot change over the years (1998-2015) several trends of changes have been identified among the districts through which the national highway, N1 has been passed. Incorporating severity index together with hotspot changes some districts have been identified as critical area for study. To identify whether there is any relation between the accident hotspot, severity and landcover the corresponding change in landcover of those districts within the same period is analysed. Categories shown in landcover are waterbody, vegetation, barren, built-up area where

Built-up Area = Urban and rural area, settlements and transportation infrastructure

Greeneries = Shrub land, rubber plantation, rained herbaceous crops, mangrove plantation, irrigated herbaceous crops, and crop in sloping land/tea, hilly forest

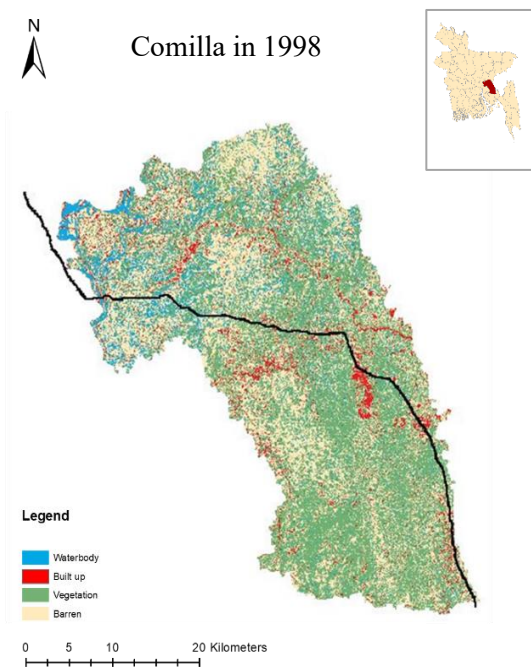
Water Body = River, reservoir/ponds, lake, canal, Bay of Bengal and low land

Bare Soil = Sand, sea -beach, open space, bare and exposed soils etc.

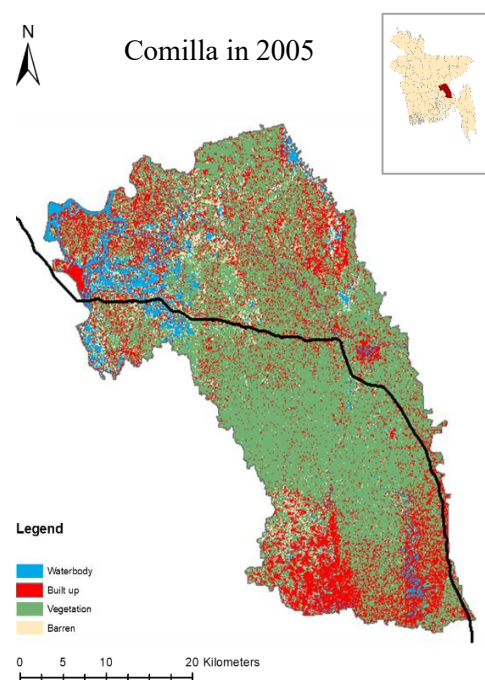
Comilla: In 1998 the amount of built-up area was about 12.72% of the total area of Comilla. Within next 5 years this amount drastically increased more than twice and reached to 30.04% in 2005. The major built up took place near the district boundary rather than near the highway which seem to be near highway in 2000. The built-up area continues to increase by the following years occupying about 35 -38% area from the year 2010 to 2015. The direction of change was noticed towards the highway from border occupying non-built-up areas like barren land, waterbody and vegetation.

Table 2: Amount of built-up area with in Comilla District

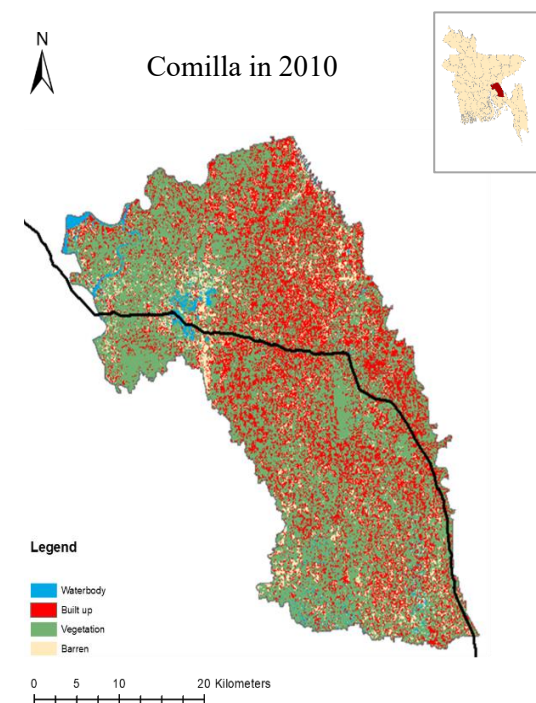
District	Built-up Area (% of total district area)			
Comilla	1998	2005	2010	2015
	12.72	30.04	35.36	38.30



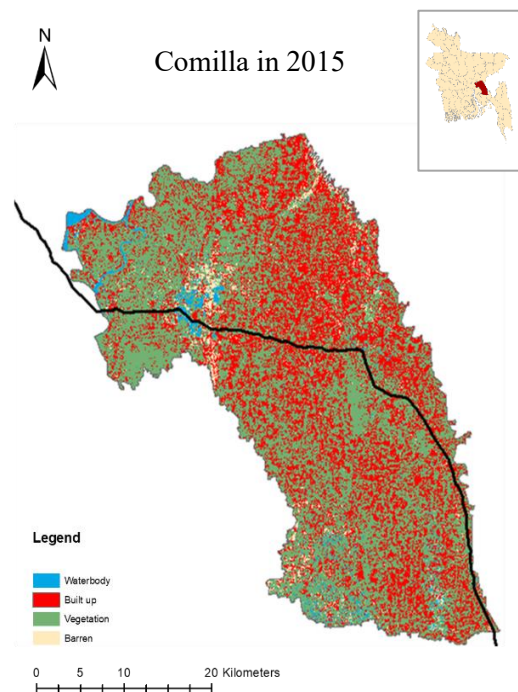
Source: Landsat 4-5, USGS



Source: Landsat 4-5, USGS



Source: Landsat 4-5, USGS



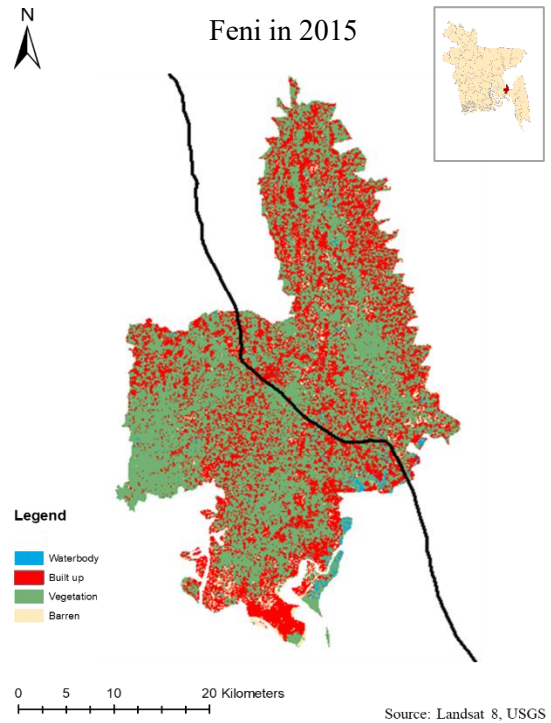
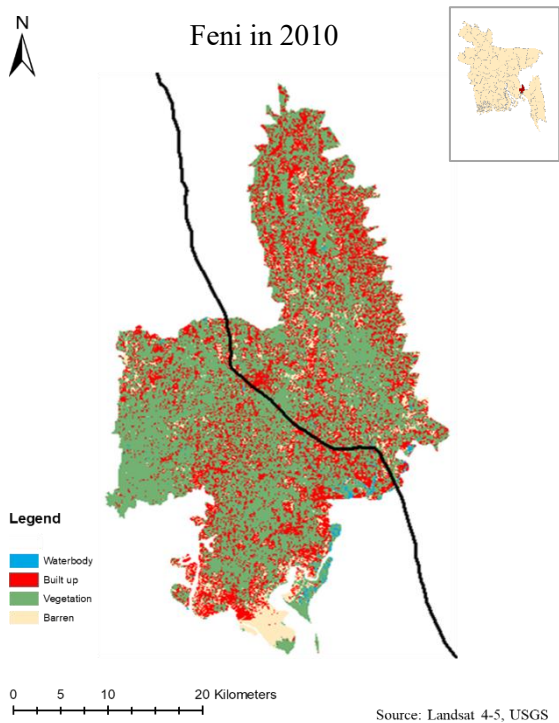
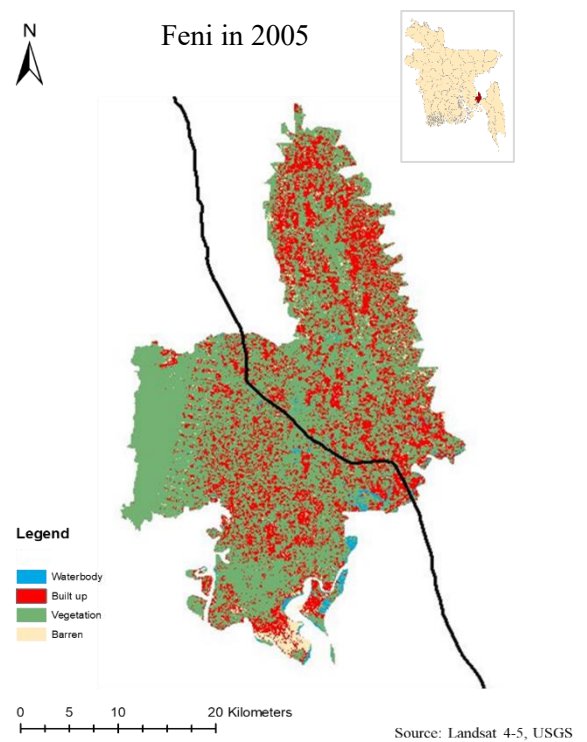
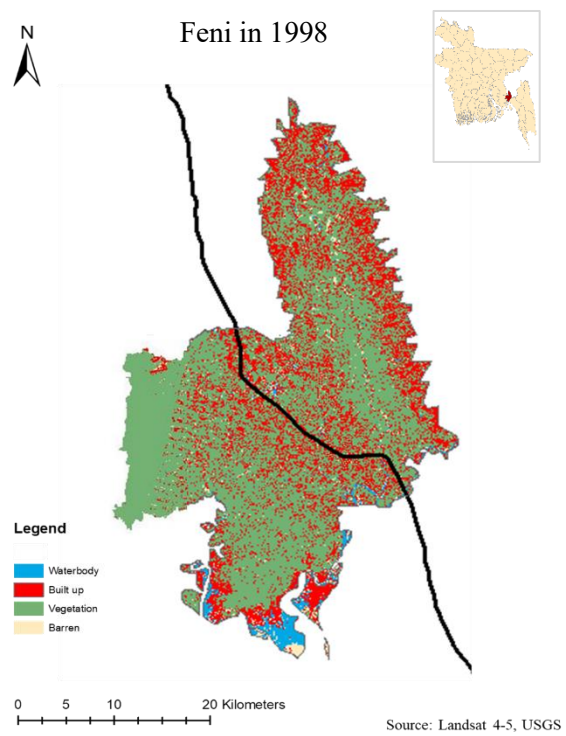
Source: Landsat 8, USGS

Map 3: Land Cover Change of Comilla 1998-2015

Feni: Built -up area seems to be higher in Feni than Comilla for the time being in fact about 28.48% of land was occupied for that purpose in 1998 among which most of the area was near the national highway. In the following years the amount of occupied land as built-up area continues to increase and reached up to 35.20% in 2015. The western part seems to develop relatively slower than the northeast portion of the district. Over the years the change was not so drastic rather gradual and seems to take place near district border.

Table 3: Amount of built-up area within Feni District

District	Built-up Area (% of total district area)			
Feni	1998	2005	2010	2015
	28.48	29.56	31.04	35.20

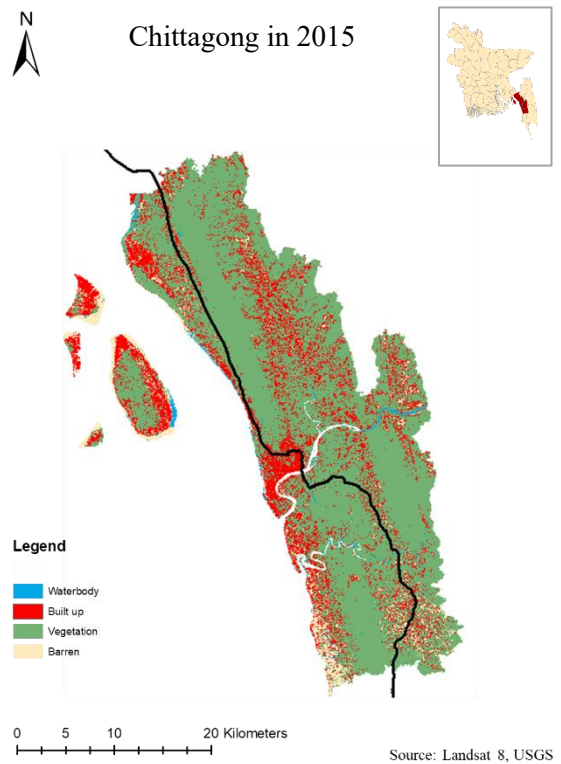
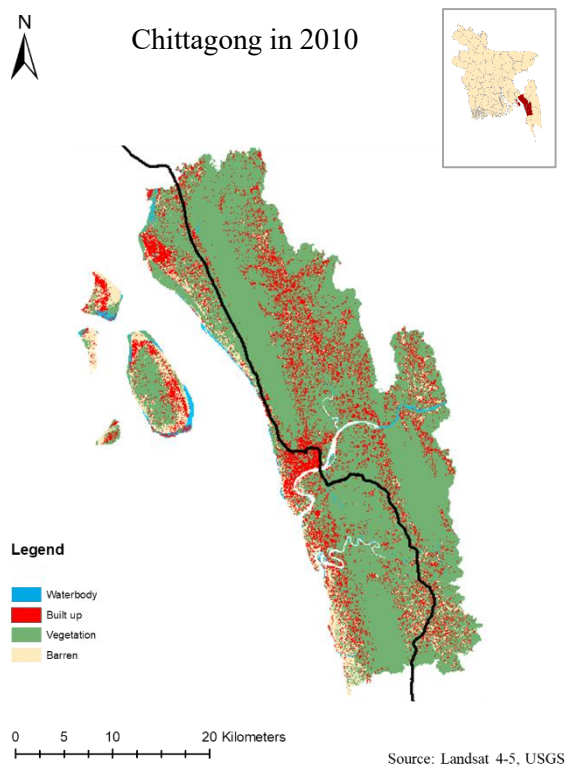
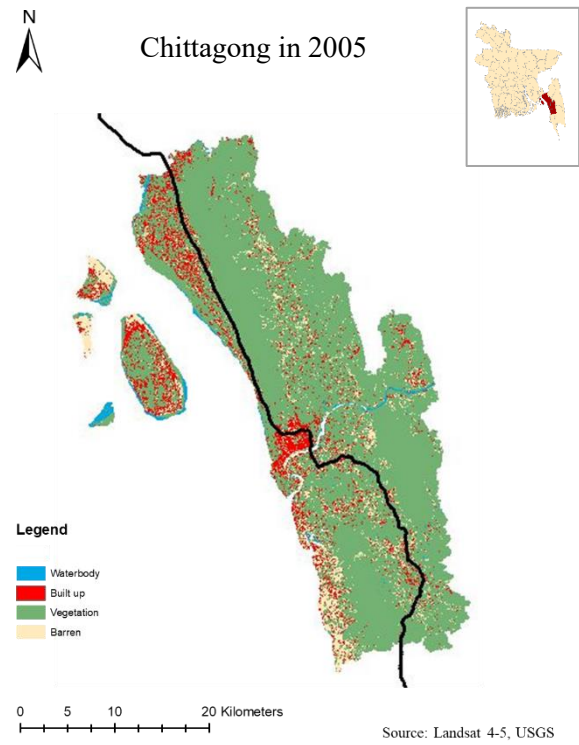
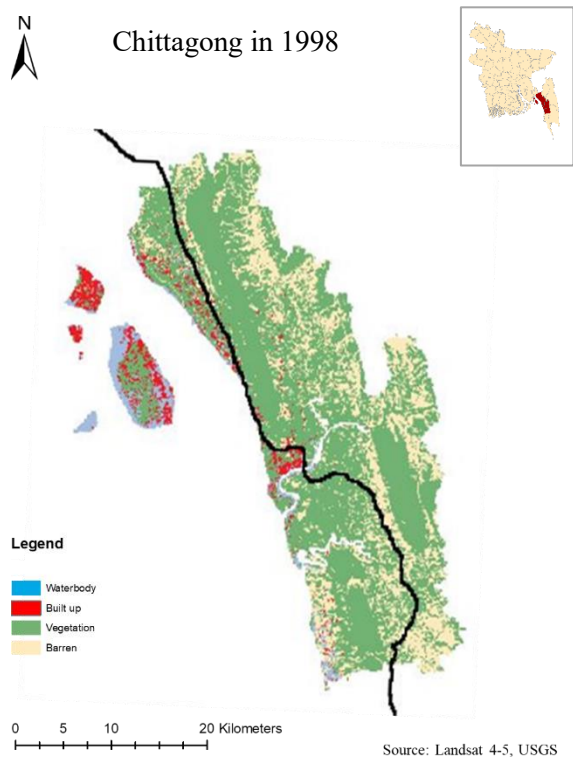


Map 4: Land Cover Change of Feni 1998-2015

Chittagong: It is basically hilly region with relatively higher elevation and steeper slope. Most of the area is hill tract and forest are hill forests. In 1998 most of the area was covered by vegetation. Around 8-9% was built up area. Within 5 years the built-up area increases very slowly and came to 10% almost. The change from 2005 to 2015 seems to be gradual and at the end of this period the figure came to 20.22%. The settlement seems to start and flourish near the bank of river Karnafuli as most of the built-up area are clustered there under Bakalia, Kotwali, kulshi Panchlaish and Chandgaon upazilla.

Table 4: Amount of built-up area within Chittagong District

District	Built-up Area (% of total district area)			
	1998	2005	2010	2015
Chittagong	8.72	10.06	16.71	20.24

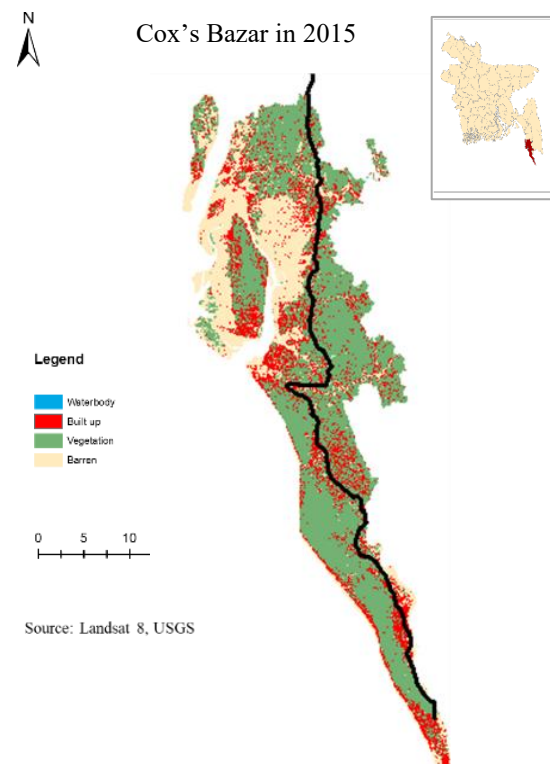
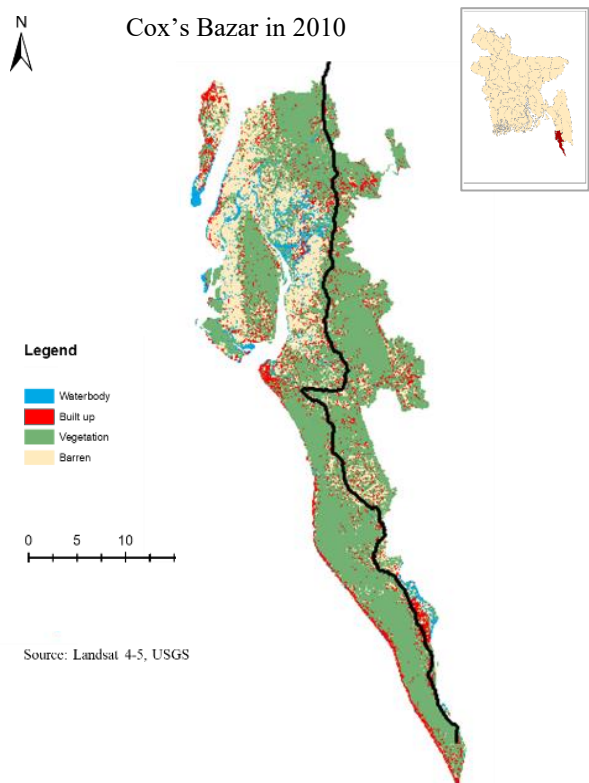
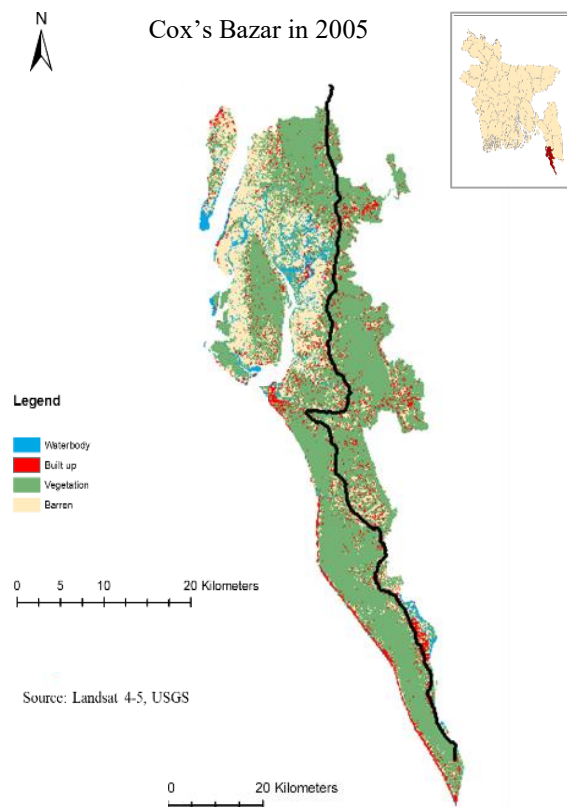
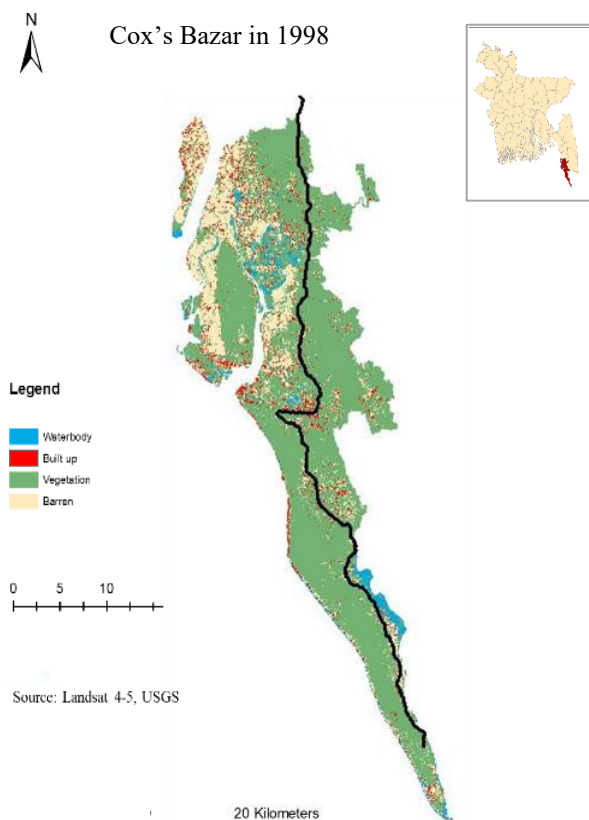


Map 5: Land Cover Change of Chittagong 1998-2015

Cox's Bazar: It was the district of least built-up area compared to the other districts included in study area. Only 5.33% of the total district was cover by built-up area in 1998. In fact, the amount was constantly low up to 2010. From 2010 to 2015, only within 5 years built-up area has drastically changed and reached to 20% in 2015 which was only 10% in 2010. The change seems to be oriented towards Moheshkhali, Naikhongchhari, Cox's bazar Sadar and Teknaf.

Table 5: Amount of built-up area within Cox's Bazar District

District	Built-up Area (% of total district area)			
	1998	2005	2010	2015
Cox's Bazar	5.33	8.45	10.37	20.22



Map 6: Land Cover Change of Cox's Bazar 1998-2015

From the percentage of built-up area for each individual district the percentage of transition from non-built up to built-up area can be calculated. These transition percentage indicate several trends for individual districts shown in figure 7. For Comilla the transition percentage

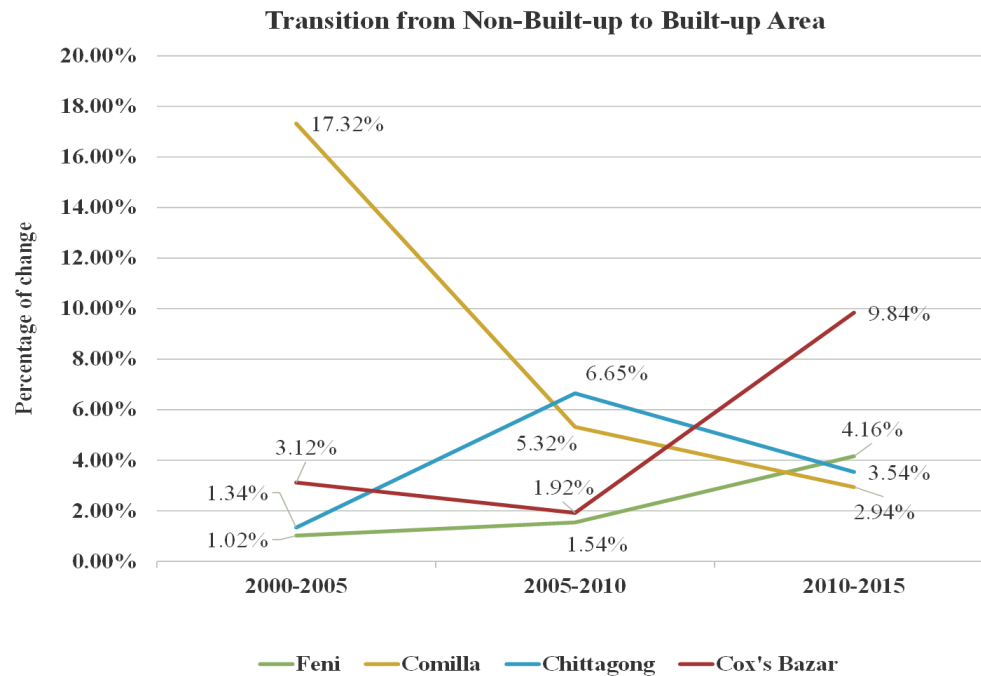


Figure 7: Trend of transition from non-built-up to built up area.

seems to decrease whereas Feni shows a increasing trend of transition. In case of Chittagong and Cox's Bazar the transition fluctuates but in the opposite direction.

Comparison of Trends and Relation Analysis:

From above analysis some trends of changes on the basis of number of accidents, severity and landcover has been identified. To analyse all those trends a cross table is given below

Table 6: Cross table showing Relation between Change Trends of Accident Hotspot, Severity and Land Cover

District	Transition Trend (Non-built up to built up)	Hotspot Changing Trend (No. of Accidents)								
		Increasing			Fluctuating			Constant		
		High	Moderate	Low	High	Moderate	Low	High	Moderate	Low
Comilla	Increasing	0	0	0	0	0	0	0	0	0
	Decreasing	1	0	0	0	0	0	0	0	0
	Fluctuating	0	0	0	0	0	0	0	0	0
Feni	Increasing	0	0	0	0	1	0	0	0	0
	Decreasing	0	0	0	0	0	0	0	0	0
	Fluctuating	0	0	0	0	0	0	0	0	0
Chittagong	Increasing	0	0	0	0	0	0	0	0	0
	Decreasing	0	0	0	0	0	0	0	0	0
	Fluctuating	0	0	0	1	0	0	0	0	0
Cox's Bazar	Increasing	0	0	0	0	0	0	0	0	0
	Decreasing	0	0	0	0	0	0	0	0	0
	Fluctuating	0	0	0	0	0	0	0	0	1

From the above table, we can compare the hotspot changing trend including severity with land cover where different district shows different association for those variables. In case of Comilla, the accident hotspot changing trend is increasing whereas landcover changing trend (transition from non-built up to built up) is decreasing where severity is high. It means that in Comilla transition from non-built up to built up area gradually decrease but still no of accidents keep on increasing. Similarly, for Feni trend of land cover change and hotspot change do not match where severity was moderate. In Feni, accident number increased upto 2010 and then fallen but non-built up area keep on converting to built up at increasing rate. Again, for Chittagong landcover change trend match to hotspot changing trend. In case of Chittagong, the accident hotspot changing trend fluctuates. Number of accident increase within the period of

2005-2010 and suddenly after that fall. Similarly transition from non-built up to built up area reaches its peak at that period and fall down after that. Accidents throughout the period were severe even though the number decrease within 2010 to 2015. Further for Cox's Bazar those trends do not match.

Findings

In the first part, we have identified some areas which are in vulnerable condition by finding out their trend in terms of hotspots for total accident and severity for those consequences. Some areas that are critical according to number of total accidents can be explained as having either serious consequences or medium. We have found 3 types of trends for both severity and accident hotspots.

- Comilla has increasing trend of total accidents and high severity. It implies that the number of total accidents had increased year by year and their consequences were also serious.
- Feni showed fluctuating trend of total accidents and the consequences of accidents were mainly moderate and serious at few areas. It implies that though the number of total accidents differed over time, their consequences were not much serious.
- Chittagong had shown the fluctuating trend in terms of total accidents and high severity. It means that though the number of total accidents differed over time, their consequences were much serious.
- Cox's Bazar was consistently in a low-risk zone of accidents and low severity. So, Cox's Bazar is the safest district with respect to both number of total accidents and severity of their consequences.

In the second part, we have explained the trend of changing land cover over years for critical areas. We have found out 3 types of trends for land cover change.

- Comilla had a decreasing trend of changing built up area. The direction of change was noticed towards the highway from border.
- Feni had an increasing trend of changing built up area near district border.
- Chittagong had fluctuating trend (increasing to decreasing rate) of changing built up area. The settlement seemed to start and flourish near the bank of river Karnafuli.

- Cox's Bazar had fluctuating trend (decreasing to increasing rate) of changing built up and seemed to flourish in the areas close to sea and river.

In general, the number of accidents can be increased with respect to increasing built up area. Built up area means concentration of activities, people and vehicle movement. So, the possibility of accidents is high if most built up areas are close to highway though severity can be low. Moreover, drivers tend to drive at a high speed in free roads and it becomes difficult to control speed all of a sudden. For this reason, generally severity of accidents should be high if most non built-up areas are close to highway though number of accidents can be low (Arnau et. al., 2020)

We have tried to establish a connection among the different trends of land cover, severity and hotspot for total accident. In case of Comilla, the accident trend is increasing which does not match with decreasing landcover changing trend (transition from non-built-up to built up) where severity is also high. Also, for Feni, trend of land cover change and hotspot change do not match where severity was moderate. Further for Cox's Bazar those trends do not match. On the other hand, Chittagong land cover change trend match to hotspot changing trend as both fluctuates. At first both the accident rate and transition from non-built up to built-up trend increases and gradually decreases over time. Also, the severity is high as the non- built up to built up change trend decreases which match our general concept.

Here, Feni, Comilla, and Cox's Bazar do not match the trends, but Chittagong precisely follow the above concept. As N1 has crossed through the main part of Chittagong city, the land cover change has a influence on accident rate and severity. So based on our study, we cannot clearly define that land cover change has a relation with the trend of hotspots and severity as we also have found exceptions. To identify exact association/ relation further analysis is required.

Conclusion

National highway 1 (N1) connects Dhaka and Chittagong which are two major cities of Bangladesh. So N1 is very important for both economy and transportation network. On the other hand, N1 is one of the most accident-prone highways in Bangladesh. Comilla, Feni and Chittagong are the most critical areas in terms of total number of accidents through which N1 has crossed. Moreover, the consequences of those accidents vary over the areas. In hotspot analysis based on total accident, increasing and constant trend for Comilla and Cox's Bazar

respectively and also fluctuating trend for Feni and Chittagong have been identified. Through analysis, severity trend of these areas is classified into three types as high, medium and low. Comilla and Chittagong are in high trend whereas Feni is in medium and Cox's Bazar is in low trend. In addition, land cover change (transition from non-built up to built up) has also showed three different trends increasing in Feni, decreasing in Comilla and fluctuation in Chittagong and Cox's Bazar. The trends of land cover change, hotspot and severity do not match for Feni, Comilla and Cox's Bazar. On contrary, these trends precisely match for Chittagong like increasing trend at initial to gradually decreasing trend in the same time period. N1 has crossed through one of the most vibrant areas of Chittagong city. So as built-up area is the main characteristics of city center, land cover has an impact on accident rate and severity. But this study cannot clearly define that land cover has a relation with accident hotspot and severity as there are some exceptions also. Further analysis is needed to identify the exact association.

References

- Anderson, T. (2007). Comparison of spatial methods for measuring road accident 'hotspots': a case study of London. *Journal of Maps*, 3(1), 55-63.
- Anderson, T. K. (2009). Kernel density estimation and K-means clustering to profile road accident hotspots. *Accident Analysis & Prevention*, 41(3), 359–364. <https://doi.org/10.1016/j.aap.2008.12.014>
- Choudhary, J., Ohri, A., & Kumar, B. (2015, December). Spatial and statistical analysis of road accidents hot spots using GIS. In 3rd Conference of Transportation Research Group of India (3rd CTRG).
- Cabrera-Arnau, C., Prieto Curiel, R., & Bishop, S. R. (2020). Uncovering the behaviour of road accidents in urban areas. *Royal Society open science*, 7(4), 191739.
- Day, K. L. (2006). Assessing the impact of highway development on land use/land cover change in Appalachian Ohio (Doctoral dissertation, Ohio University).

Dereli, M. A., and S. Erdogan. 2017. "A New Model for Determining the Traffic Accident Black Spots Using GIS-aided Spatial Statistical Methods." *Transportation Research Part A: Policy and Practice* 103: 106–117.

Erdogan, S., Yilmaz, I., Baybura, T., & Gullu, M. (2008). Geographical information systems aided traffic accident analysis system case study: city of Afyonkarahisar. *Accident Analysis & Prevention*, 40(1), 174-181. Manap, N., Borhan, M. N., Yazid, M. R. M., Hambali, M. K. A., & Rohan, A. (2019).

Ewing, R., Schieber, R. A., & Zegeer, C. V. (2013). Urban sprawl as a risk factor in motor vehicle crashes. *Transportation Research Part A: Policy and Practice*, 50, 62-70.

Feng, S., Liu, S., Jing, L., Zhu, Y., Yan, W., Jiang, B., ... & Jia, J. (2021). Quantification of the

FlahautB., Mouchart, M., Martin, E. S., & Thomas, I. (2003). The local spatial autocorrelation and the kernel method for identifying black zones. *Accident Analysis & Prevention*, 35(6), 991–1004. [https://doi.org/10.1016/s0001-4575\(02\)00107-0](https://doi.org/10.1016/s0001-4575(02)00107-0)

Harirforoush, H., and L. Bellalite. 2016. "A New Integrated GIS-based Analysis to Detect Hotspots: A Case Study of the City of Sherbrooke." *Accident Analysis & Prevention* (Advance online publication). doi:10.1016/j.aap.2016.08.015.

ITF (International Transport Forum). 2017. *Road Safety Annual Report 2017*. Paris: OECD Publishing. doi:10.1787/irtad-2017-en.

Jones, A. P., Langford, I. H., & Bentham, G. (1996). The application of K-function analysis to the geographical distribution of road traffic accident outcomes in Norfolk, England. *Social Science & Medicine*, 42(6), 879-885.

Lambin, E. F., Turner, B. L., Geist, H. J., Agbola, S. B., Angelsen, A., Bruce, J. W., ... & Xu, J. (2001). The causes of land-use and land-cover change: moving beyond the myths. *Global environmental change*, 11(4), 261-269.

Lein, J. K., & Day, K. L. (2008). Assessing the growth-inducing impact of the Appalachian Development Highway System in southern Ohio: Did policy promote change?. *Land Use Policy*, 25(4), 523-532.

Le, K. G., Liu, P., & Lin, L. T. (2020). Determining the road traffic accident hotspots using GIS-based temporal-spatial statistical analytic techniques in Hanoi, Vietnam. *Geo-spatial Information Science*, 23(2), 153-164.

Saleh, S., & Al Islam, S. B. Safety Investigation and Assessment of Dhaka-Chittagong National Highway in Bangladesh Sensing, 13(7), 1340.

Shariff, S. R., Maad, H. A., Halim, N. N. A., & Derasit, Z. (2018). Determining hotspots of road accidents using spatial analysis. Indonesian Journal of Electrical Engineering and Computer Science, 9(1), 146-151.

Steenberghen, T., Dufays, T., Thomas, I., & Flahaut, B. (2004). Intra-urban location and clustering of road accidents using GIS: a Belgian example. International Journal of Geographical Information Science, 18(2), 169-181.

Verburg, Peter & Kok, Kasper & Pontius, Robert & Veldkamp, Antonie. (2008). Modeling Land-Use and Land-Cover Change. 10.1007/3-540-32202-7_5.

World Health Organization. (2013). Global status report on road safety 2013: supporting a decade of action: summary (No. WHO. NMH. VIP 13.01). World Health Organization.

Appendix

Model of Hotspot Analysis:

For conducting Kernel Density estimation, an inbuilt tool, Kernel Density is used in GIS. As input the shape file of road layer for which hotspot is to be identified is inserted. To define the masking area a 1km buffer layer from the same road shape file is created using buffer tool under geoprocessing tool. The following data model shows the procedure of work.

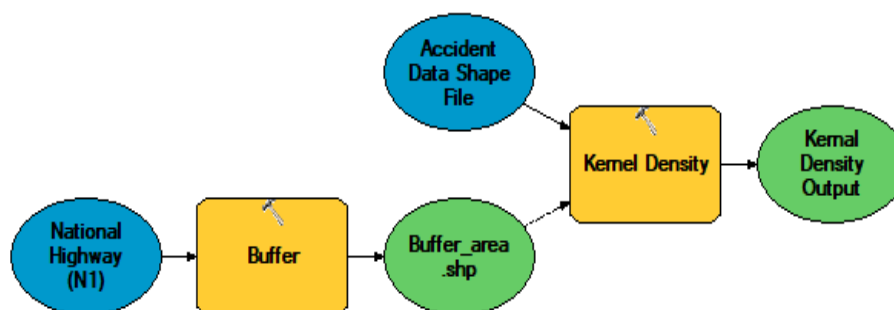


Figure 8: Procedure of Hotspot analysis in GIS using

Land Cover Map Preparation:

Preparing land cover for a small scale that means for large area is very much challenging as reflectance value for barren land and built-up area is closer. In general, to identify built up area using NDBI is used in most of the study. In NDBI, positive values suggest built up and negative value implies no-built up area. But unfortunately, due to some unknown error after reclassification most of the area shows as built up which is quite absurd.

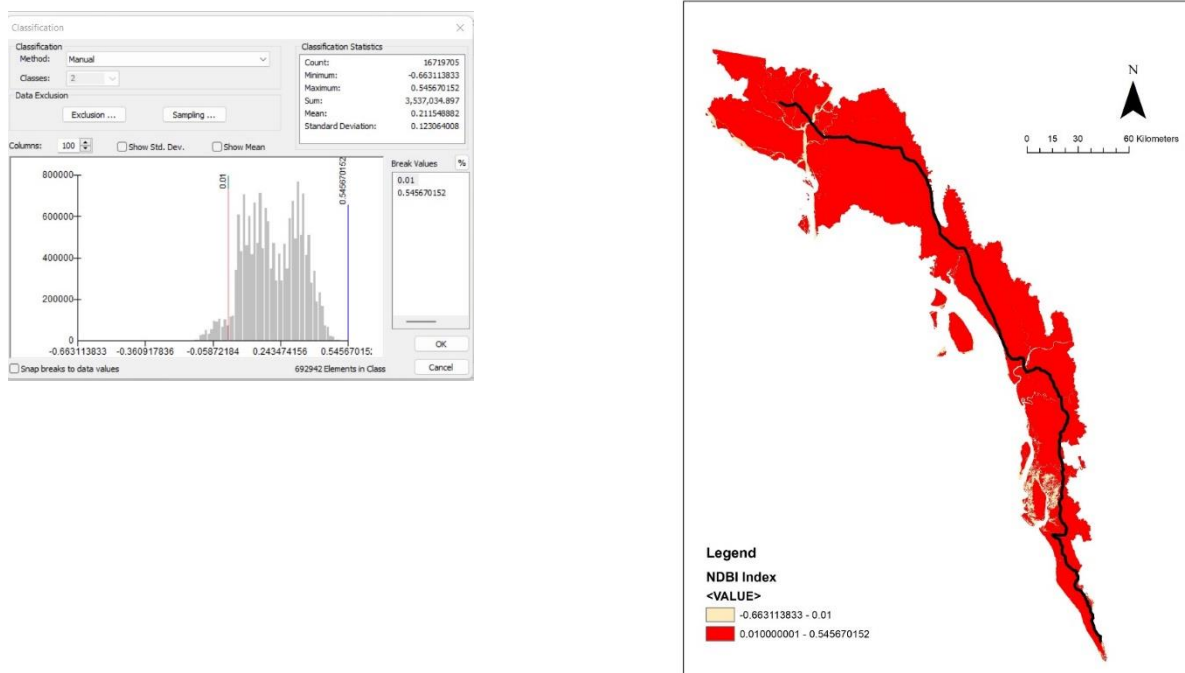


Figure 9: Error in NDBI Map

Ultimately landcover map is prepared through supervised classification using maximum likelihood method. For that, the raster tiles of different bands are collected from Landsat and make composite band for each of the tile covering the study area. All the composite tiles are combined through mosaic and clipped according to the boundary of study area. Then the clipped raster file is trained considering training sample more than 60 and create signature file. And finally using clip raster and signature file as input we get the final land cover map.

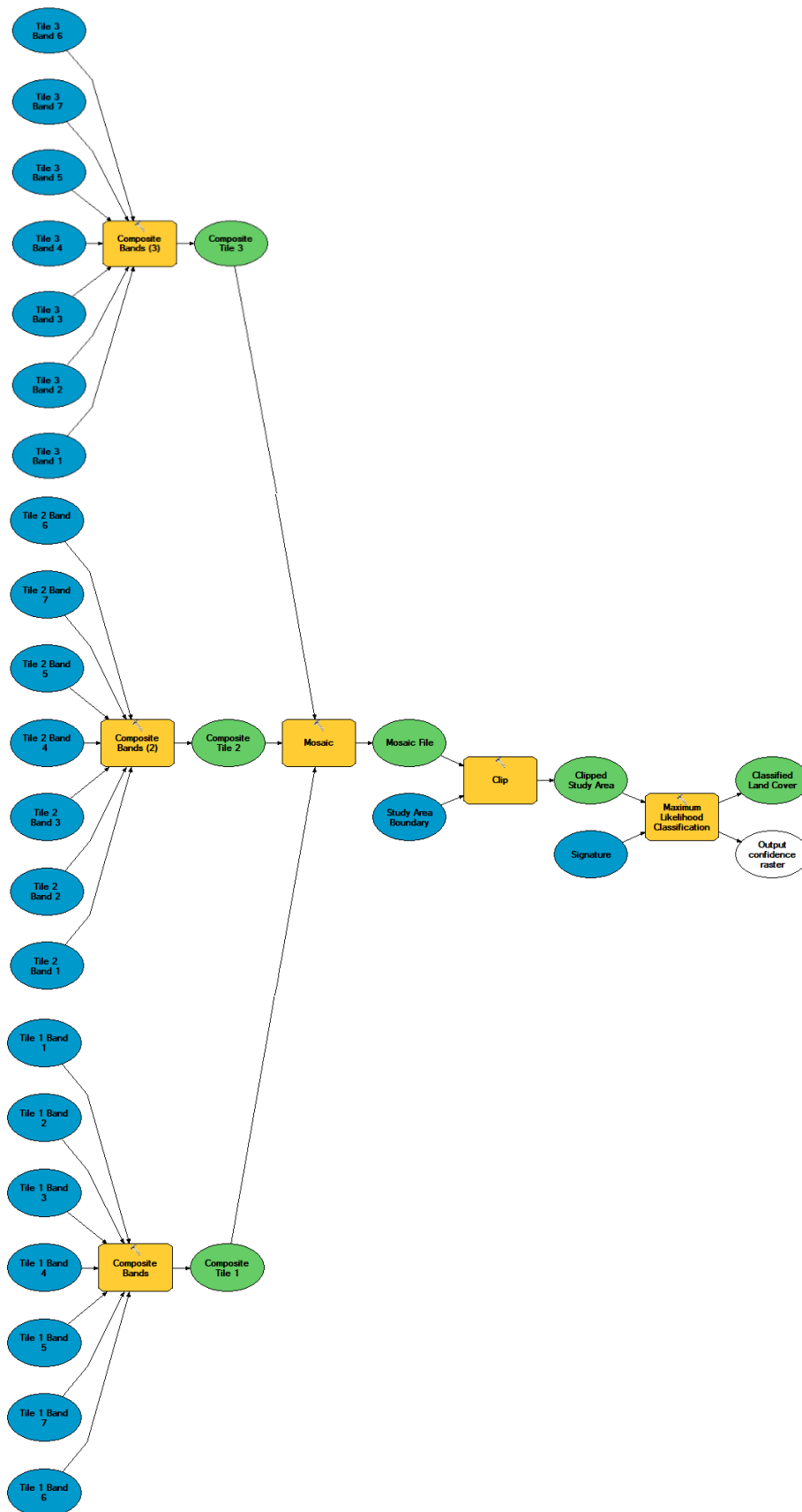


Figure 10: Procedure of Land Cover Map Preparation in GIS using Supervised Classification